

Honeywell

WebGUS
Authoring and Runtime
Guide

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TotalPlant

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About This Document

This document describes,

- The concept of GUS displays inside of WebGUS
- How to script ActiveX WebGUS controls,
- How to use WebGUS to run a GUS display

Release Information

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Symbol Definitions

The following table lists those symbols used in this document to denote certain conditions.

Symbol	Definition
	ATTENTION: Identifies information that requires special consideration.
	TIP: Identifies advice or hints for the user, often in terms of performing a task.
CAUTION	Indicates a situation which, if not avoided, may result in equipment or work (data) on the system being damaged or lost, or may result in the inability to properly operate the process.

Symbol Definitions

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1. GUS Displays in WebGUS

1.1 About the WebGUS Authoring and Runtime Guide

Who should use this Authoring Guide

Individuals who need to create web pages containing GUS displays that will be presented by WebGUS need to be familiar with the information in this guide.

A user of WebGUS should:

- Have a basic understanding of HTML syntax.
- Understand how to do web page scripting using VBScript or JavaScript.

What's needed to support authoring

See *WebGUS Requirements and Resources* in the *WebGUS Planning and Installation Guide* for requirements to author and display the WebGUS component.

Additional documentation references

Books that have been referenced in this manual or may be useful while using WebGUS are:

- *Display Authoring Tutorial*
- *Display Builder User's Guide*
- *Display Scripting User's Guide*
- *GUS HCI/OPC Data Access User's Guide*
- *TPS Acronyms & Abbreviations*
- *TPS Terms & Definitions*

1.2 WebGUS Functional Overview

Role of WebGUS ActiveX control

The WebGUS product brings operational control room GUS displays to the supervisory desktop PC, via an ActiveX control. This ActiveX control encapsulates the GUS runtime functionality, resulting in no required changes to existing libraries of operational displays.

It thus opens the availability of GUS display functionality to any network-connected computer with the appropriate software configuration and security access rights.

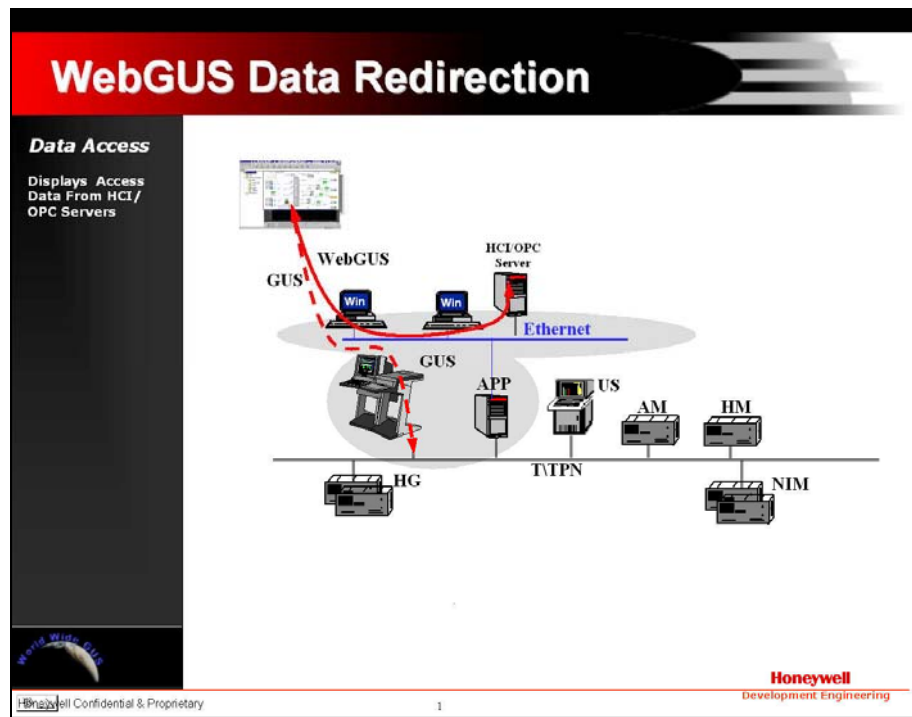


ATTENTION

The WebGUS control is not intended for primary operator control. It does not provide the same degree of alarm handling and performance.

Role of WebGUS Redirection

WebGUS contains a data access redirection feature. This redirection feature allows existing GUS display data references to be redirected to one or more “surrogate” HCI/OPC data servers.



Display Data Redirection Example

WebGUS in Control Room Displays

While the WebGUS product is primarily intended to bring control room displays out to the desktop, the software may provide benefits to the control room – that is, to GUS nodes. Because GUS displays are themselves ActiveX containers, a GUS node can leverage the WebGUS ActiveX control to encapsulate separate GUS displays into other “main” GUS displays.

This method of encapsulating GUS displays (display source) directly into “main” display has the potential benefit of emulating the “overlay” feature of the Universal Station.



ATTENTION

It is important to note however that WebGUS should not be considered interchangeable with embedded displays as described in the Display Builder User's Guide, and the Display Scripting User's Guide.

Desktop GUS Display Building

The GUS Display Builder product may now be installed on a desktop. When used in conjunction with the WebGUS software, this allows WebGUS displays to be developed, tested, and executed within the desktop environment. Others, as WebGUS displays and/or as normal GUS displays, may then make such displays available for subsequent use. For example;

- A plant engineer may use the GUS display builder on the desktop to build GUS displays that use only HCI/OPC data references or,
- A plant engineer may develop GUS displays at his or her desktop, including LCN data references, test them using the WebGUS product's data redirection functionality, and then deploy them, after validating against the LCN, to the control room.

Once deployed to the control room in standard GUS nodes, the LCN references built into the displays work as normal.

WebGUS ActiveX Containers

The WebGUS ActiveX control can be hosted by either Internet Explorer or a GUS display.

1.3 How Do You Want to Present Displays Using WebGUS

How to present displays using WebGUS

The table below lists the four major approaches to providing WebGUS displays.

If you want to ...	then see ...
configure a plant's currently used GUS displays to personnel operating from a desktop using the WebGUS product,	Tasks that an Administrator would do Integrating WebGUS ActiveX control in a web page.
author web pages to integrate GUS displays within a web page using events available with the WebGUS control so that a GUS display can interact within the web page (or other ActiveX container) using the WebGUS product,	Integrating GUS displays using WebGUS in web page
view GUS displays from a desktop using a system that already has been configured with the WebGUS product,	User Interface to GUS displays using WebGUS
present a GUS display in another GUS display using the WebGUS ActiveX control. "overlay"	Integrating GUS display in a GUS display using WebGUS

1 GUS Displays in WebGUS

1.3 How Do You Want to Present Displays Using WebGUS

2. Inserting WebGUS ActiveX Controls

2.1 Inserting WebGUS ActiveX Control into web page



ATTENTION

You must first install and register the WebGUS ActiveX Control on your system before the control can be used to display GUS displays. See Using the Environment Selection Section in the WebGUS Planning and Installation Guide.

Adding the WebGUS control to your design environment

In the following procedure, MS FrontPage 2000 was used as the design environment



TIP

The first time that MS FrontPage 2000 and the WebGUS ActiveX control are installed together on a PC, you may have to select the Customize option under The FrontPage 2000 'Add ActiveX Control', in order to add the Honeywell WebGUS ActiveX control.



ATTENTION

If an .htm page that contains more than one WebGUS control is opened within FrontPage, viewing it in the "Preview" mode will raise a message box "Could not connect to GUS Display Runtime. Design mode assumed."

To test pages containing multiple WebGUS controls, it is necessary to save them and invoke them outside of FrontPage.

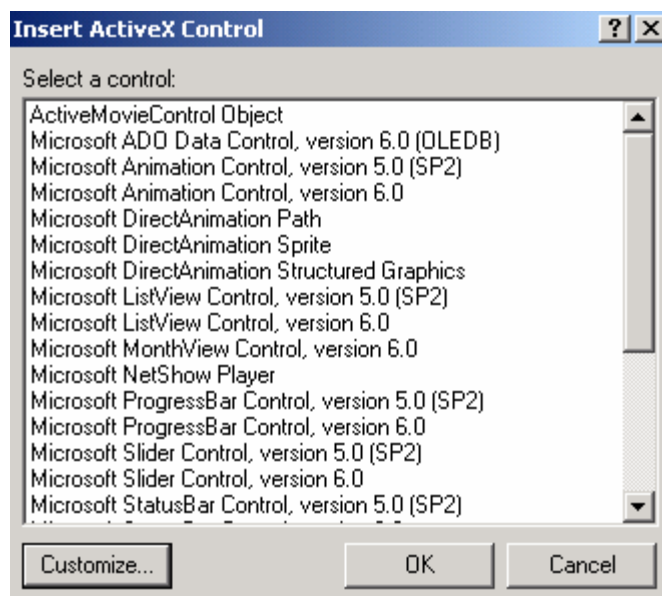
Follow the steps in the table below to insert a Honeywell WebGUS ActiveX control

Step	Action
1	Invoke MS FrontPage 2000.
2	Select menu item Insert>Advanced>ActiveX Control

2 Inserting WebGUS ActiveX Controls

2.1 Inserting WebGUS ActiveX Control into web page

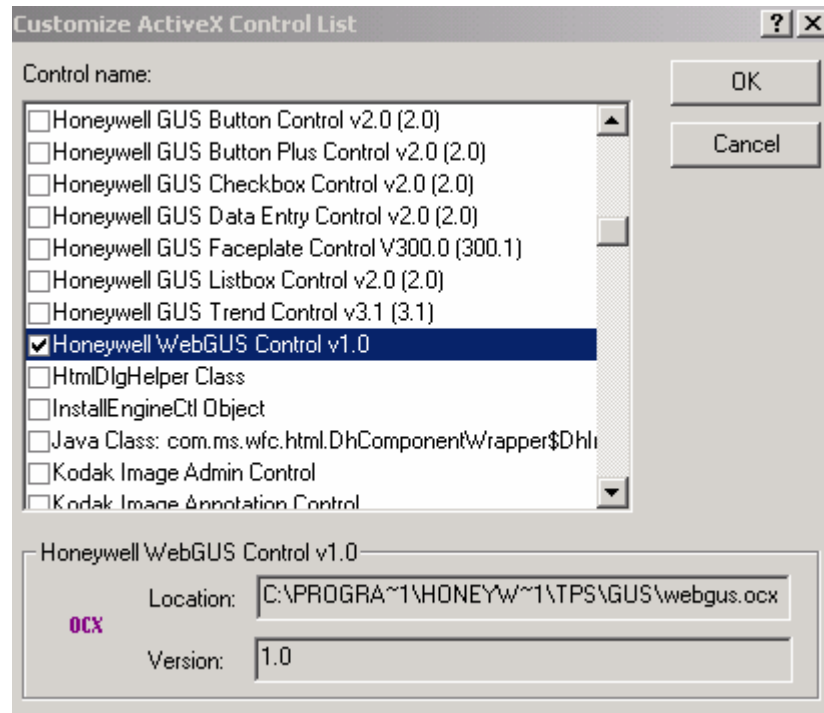
Step	Action
3	If the Honeywell WebGUS Control is not on the control list, click Customize from the Insert ActiveX Control option. If it is on the list, go to Step 5.



2 Inserting WebGUS ActiveX Controls

2.1 Inserting WebGUS ActiveX Control into web page

Step	Action
4	Select Honeywell WebGUS Control and click OK

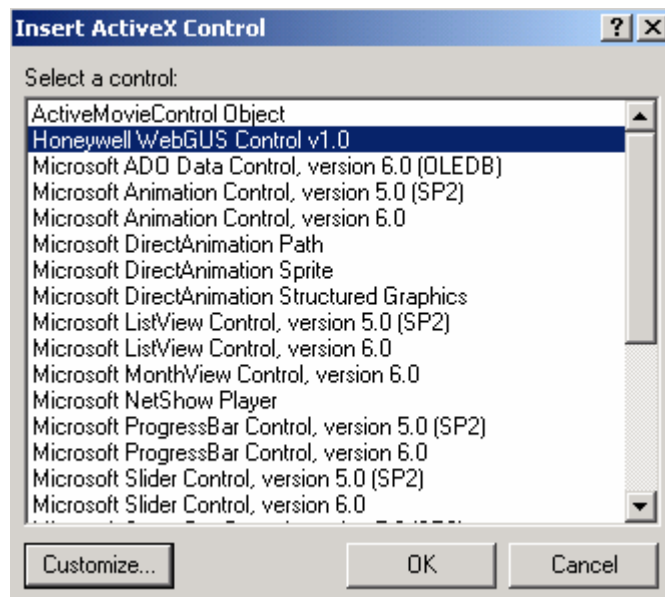


Note: The Location shown in the example above may vary by environment.

2 Inserting WebGUS ActiveX Controls

2.1 Inserting WebGUS ActiveX Control into web page

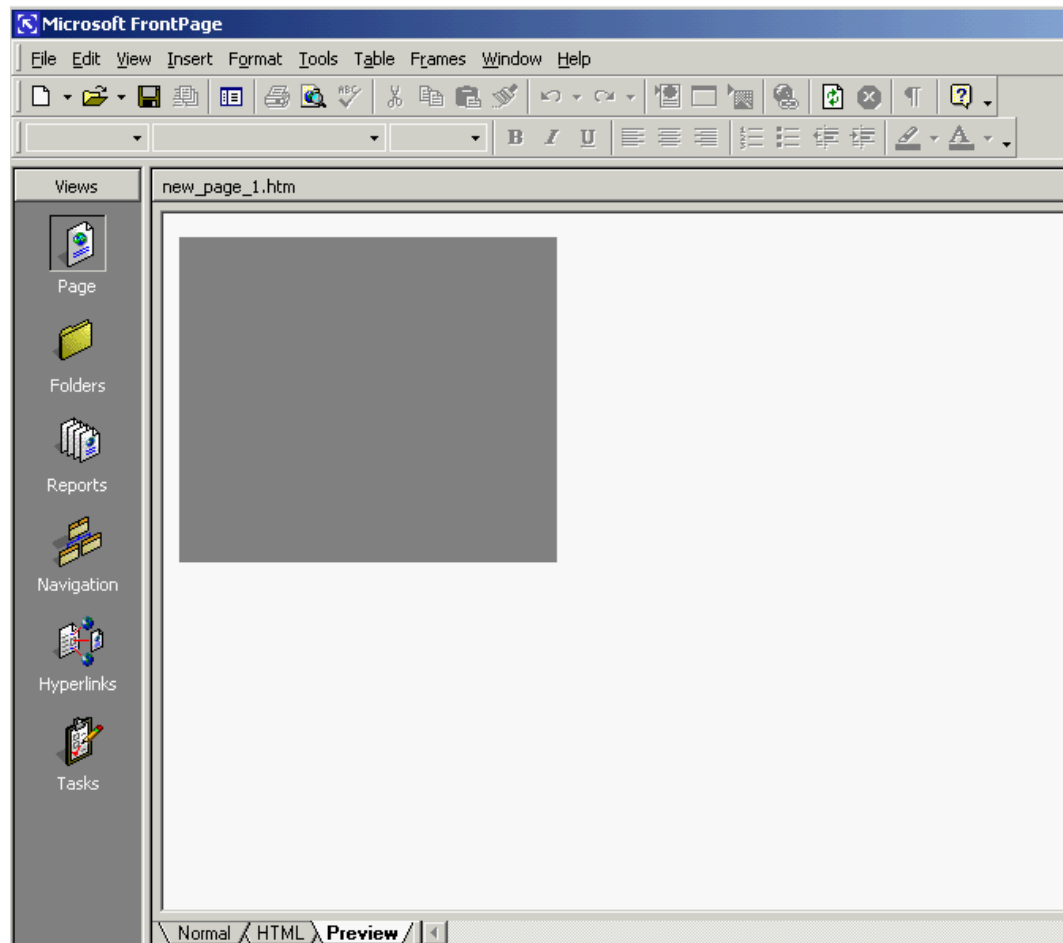
Step	Action
5	From the Insert ActiveX Control option, select Honeywell WebGUS Control and click OK .



2 Inserting WebGUS ActiveX Controls

2.1 Inserting WebGUS ActiveX Control into web page

WebGUS in a Front Page Environment



2 Inserting WebGUS ActiveX Controls

2.1 Inserting WebGUS ActiveX Control into web page

Invoking WebGUS Property Page

ActiveX containers such as FrontPage 2000 may provide their own human interface for accessing ActiveX control properties such as those of the WebGUS ActiveX control. Shown below is the FrontPage 2000 human interface for accessing the WebGUS control's properties.

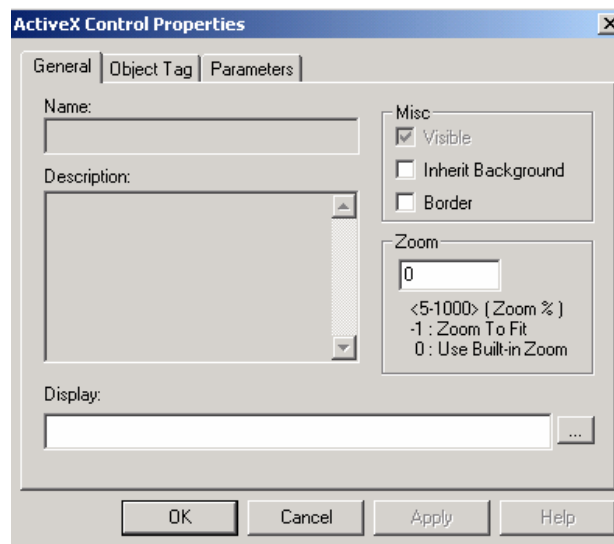


TIP

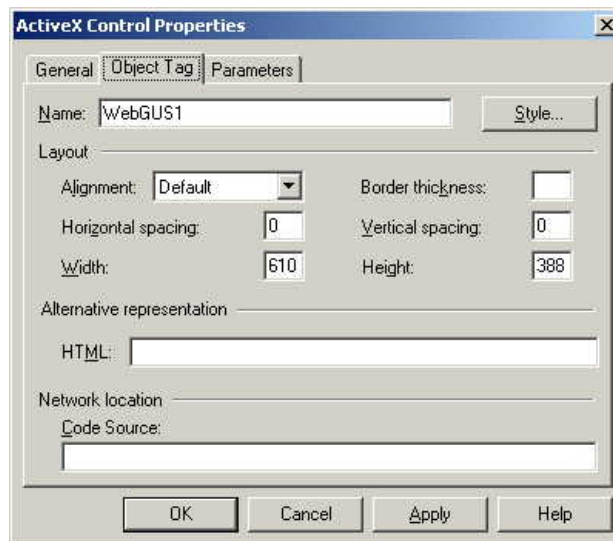
This human interface provides the same properties dialog as does the GUS container, but this dialog includes two additional tabs: Object Tag and Parameters.

For specific information on the components of these human interface features, please refer to the FrontPage 2000 user documentation.

FrontPage 2000 WebGUS Property Page - General



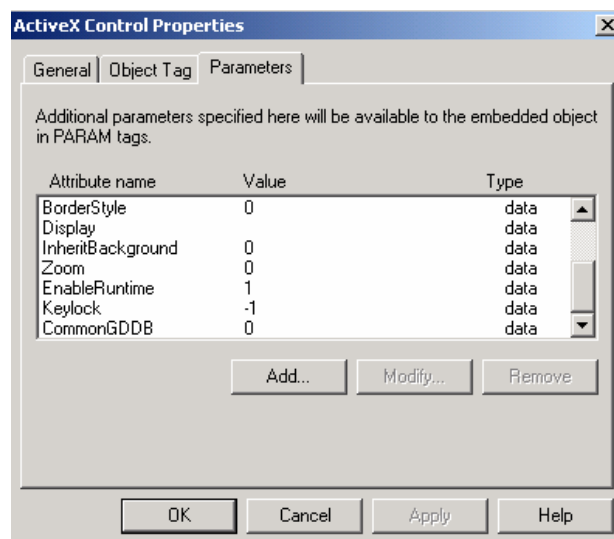
FrontPage 2000 WebGUS Property Page - Object Tag



2 Inserting WebGUS ActiveX Controls

2.1 Inserting WebGUS ActiveX Control into web page

FrontPage 2000 WebGUS Property Page - Parameter



ATTENTION

This tab shows the results of the generated web page properties within the HTML code.

You would not normally use this tab for configuring your parameters. This tab should be used for parameter viewing only.

Defining WebGUS Control Properties on the Properties Page – General

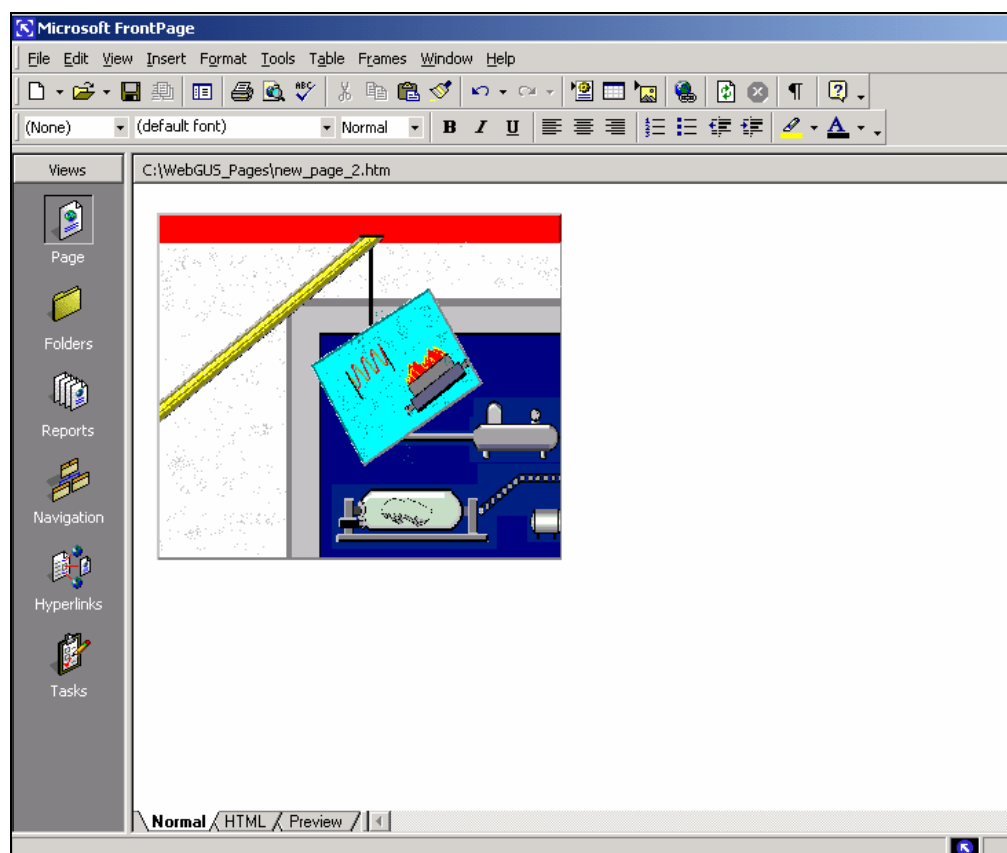
The following table lists the properties that are accessible from the WebGUS ActiveX Control Property page, General tab, with links to the WebGUS ActiveX properties table.

To View this Property Description...	See ...
Name	The object's name, if embedded in a GUS container. Used to access this instance for scripting purposes. The property UI is grayed if the property page is invoked by a different build-time container (e.g. FrontPage). A non-GUS container has other means of defining object's name.
Description	Description under Scripting WebGUS ActiveX Control Properties.
Display	Display under Scripting WebGUS ActiveX Control Properties.
Visible	Visible under Scripting WebGUS ActiveX Control Properties.
Inherit Background	InheritBackground under Scripting WebGUS ActiveX Control Properties.
Border	BorderStyle under Scripting WebGUS ActiveX Control Properties.
Zoom	Zoom under Scripting WebGUS ActiveX Control Properties.

2 Inserting WebGUS ActiveX Controls

2.1 Inserting WebGUS ActiveX Control into web page

Front Page 2000 Development Environment



ATTENTION

If a web page that contains more than one WebGUS control is opened within FrontPage, viewing it in the "Preview" mode will raise the message box,

"Could not connect to GUS Display Runtime. Design mode assumed."

To test pages containing multiple WebGUS controls, it is necessary to save them and invoke them outside of FrontPage.

2.2 Inserting WebGUS ActiveX Control into GUS Display



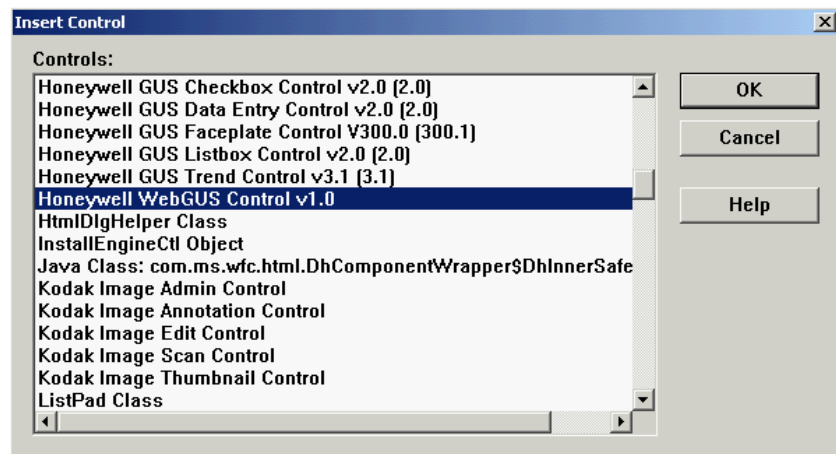
ATTENTION

You must first install and register the WebGUS ActiveX Control on your system before the control can be used to display GUS displays. See Using the Environment Selection Section in the WebGUS Planning and Installation Guide.

Insert WebGUS ActiveX Control procedure

Follow the steps in the table below to insert a WebGUS ActiveX Control onto your GUS display.

Step	Action
1	From GUS Display Builder Insert menu, select Controls.
2	Scroll and select the control Honeywell WebGUS Control Vx.x , and click OK .

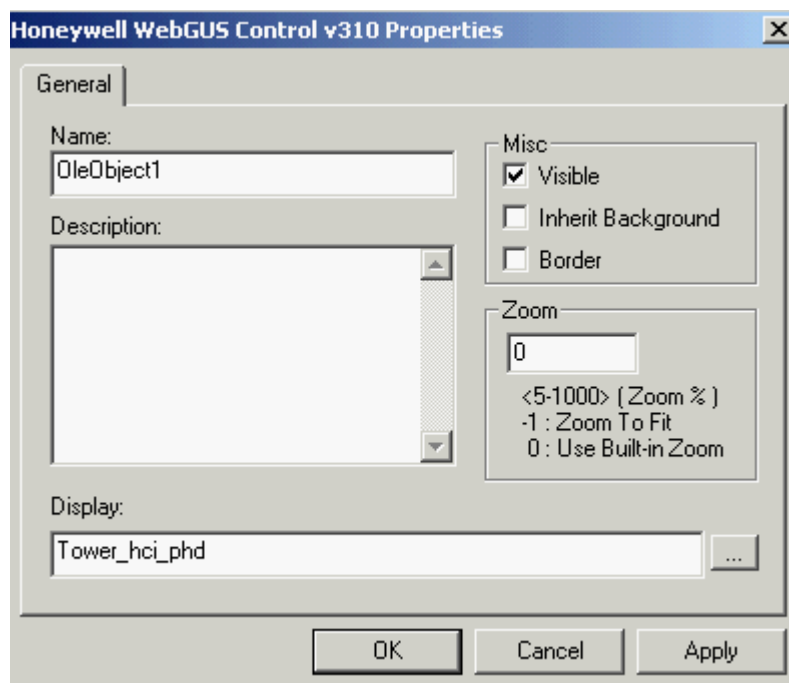


3	Hold down the left mouse button and size the control on your GUS display.
4	Double-click on the control to expose the control property page.

2 Inserting WebGUS ActiveX Controls

2.2 Inserting WebGUS ActiveX Control into GUS Display

Step	Action
5	Type in the name of the GUS display that you want to be hosted by the WebGUS control. In this example, we used Tower_hci_phd. See Invoking WebGUS Property Page.

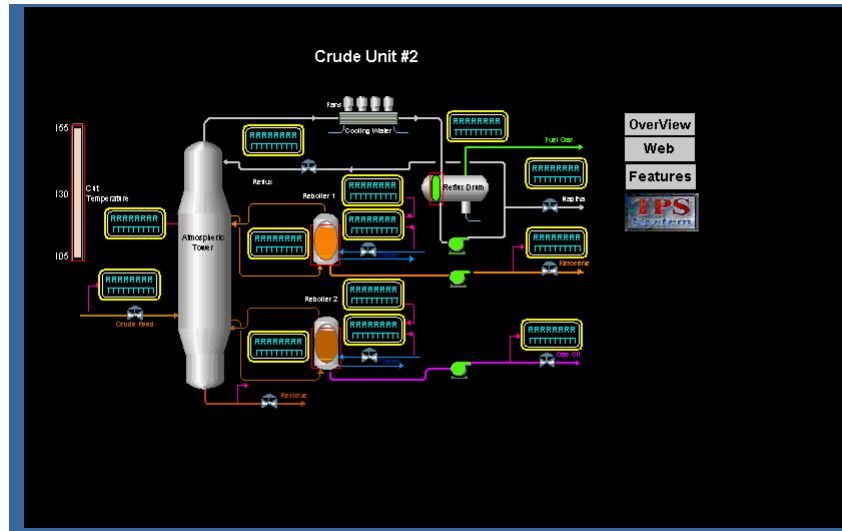


Note: The display must be in one of the folders in the configure display search path.

2 Inserting WebGUS ActiveX Controls

2.2 Inserting WebGUS ActiveX Control into GUS Display

Step	Action
6	Run the GUS display. In our example, the GUS display Tower_hci_phd.pct is shown within the WebGUS control.



7	Save the display.
---	-------------------

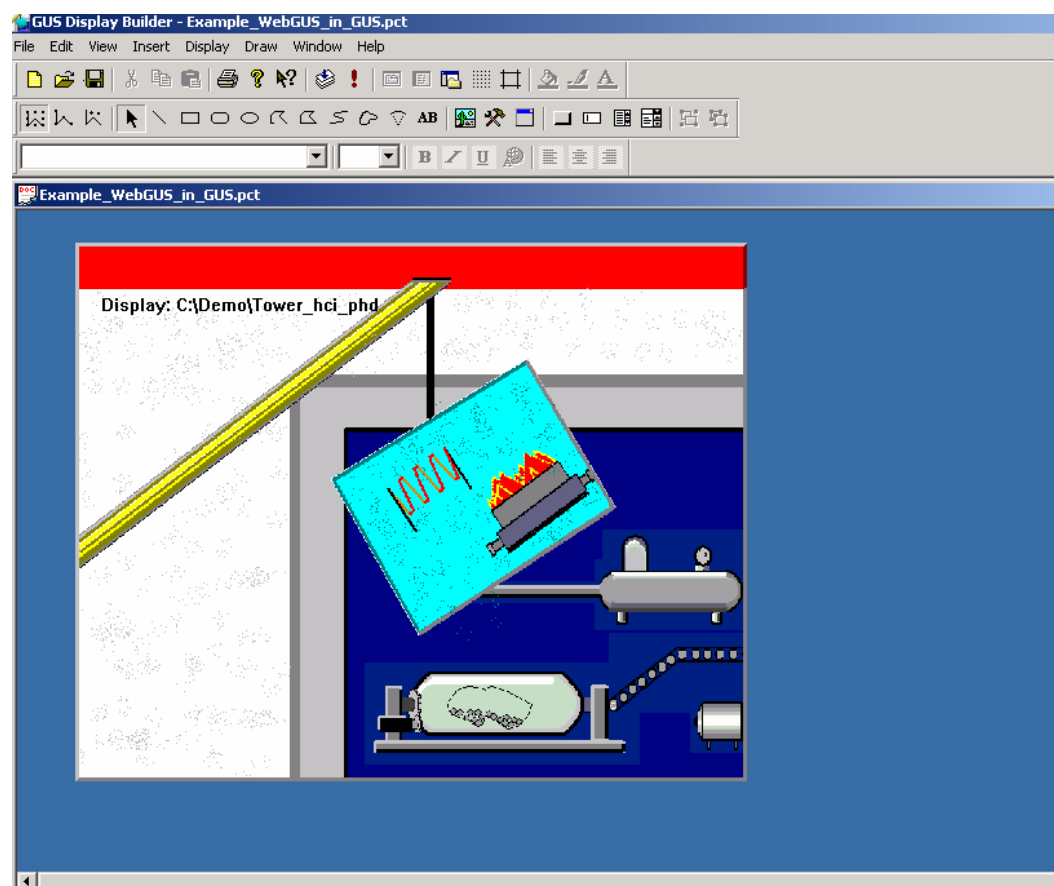
2 Inserting WebGUS ActiveX Controls

2.2 Inserting WebGUS ActiveX Control into GUS Display

WebGUS control build time example

The illustration below shows the WebGUS control that was constructed in the Insert WebGUS ActiveX Control procedure during build time, i.e. before the display is run.

WebGUS Control Build Time Example



Invoking WebGUS Property Page

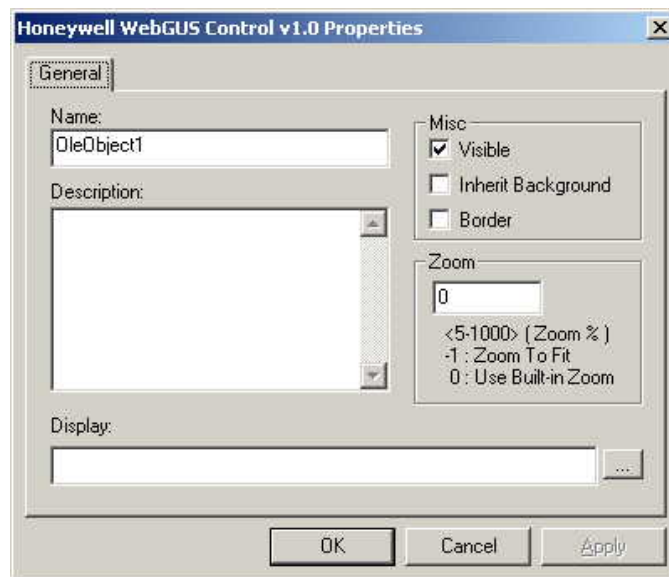
The WebGUS ActiveX control Property Page provides a build mode (only) user interface for accessing and setting WebGUS properties. Note that additional properties are available, beyond those shown by the Properties Dialog. A complete list of available properties can be found under Scripting WebGUS ActiveX Control Properties. The Properties Dialog is accessed in the standard method used by the given container for invoking an ActiveX control's Property Page. In the GUS container, this includes

- Double-clicking the control, or
- Selecting “Edit/Properties” from the main menu, while a WebGUS ActiveX control is selected.

In FrontPage 2000 and Visual Basic, the WebGUS ActiveX control Property Page is presented after the user right-mouse clicks on the control, and then selects the “Properties” menu option.

Shown below is the WebGUS ActiveX control's Property Page, with its initial default values. The object name default will vary with the given container. Shown is the default name for the first such object inserted into a GUS container display.

WebGUS Property Page From Within GUS Display Builder



2 Inserting WebGUS ActiveX Controls

2.2 Inserting WebGUS ActiveX Control into GUS Display

Defining WebGUS Control Properties on the Properties Page

The following table lists the properties that are accessible from the WebGUS ActiveX Control Property page with links to the WebGUS ActiveX properties table.

To View this Property Description...	See ...
Name	The object's name, if embedded in a GUS container. Used to access this instance for scripting purposes. The property UI is grayed if the property page is invoked by a different build-time container (e.g. FrontPage). A non-GUS container has other means of defining object's name.
Description	Description under Scripting WebGUS ActiveX Control Properties.
Display	Display under Scripting WebGUS ActiveX Control Properties.
Visible	Visible under Scripting WebGUS ActiveX Control Properties.
Inherit Background	InheritBackground under Scripting WebGUS ActiveX Control Properties.
Border	BorderStyle under Scripting WebGUS ActiveX Control Properties.
Zoom	Zoom under Scripting WebGUS ActiveX Control Properties.

3. Scripting WebGUS ActiveX Controls

3.1 Introduction to WebGUS ActiveX Control Scripting

Properties, Methods, Events

WebGUS controls expose Properties, Methods, and Events that can be scripted using available standard scripting/development environments. Below is a list of some example scripting and development environments.

- Microsoft's FrontPage 2000
- Microsoft's InterDev
- Honeywell's GUS Display Builder
- Microsoft's 2000 Office Applications (Word, Excel, PowerPoint, Access).
- Microsoft's Visual Basic
- Visual Basic and JavaScript Environments
- HTML editors (even Notepad.exe)

Properties are the characteristics of an object such as size, position, color, or text font. Properties determine the appearance and sometimes the behavior of an object.

Methods are routines provided within the object that may be called within a script, which define how the object will work with information, and respond to certain events. For example, when the Refresh() method is executed within a script, the GUS display will be re-painted on the screen.

Events are actions raised by object that may then be processed in the scripting environment. For example, ActiveX control raises the *OnStandardEvent* event when a GUS standard mouse-event [for example, OnLButtonDown] method would normally execute, allowing the WebGUS event to preempt the standard OnLButtonDown processing

3 Scripting WebGUS ActiveX Controls

3.1 Introduction to WebGUS ActiveX Control Scripting

Container script example

The initial display presented is determined by the value of the display. If the display property is not set, or if the display value is not valid, no display is presented. Container resident scripts may change the display presented by a WebGUS control by changing/setting the display property to the name of a display. Setting the display property to an empty (null) string results in no display, clearing any existing display.

When using a browser container such as Microsoft Internet Explorer, The HTML is either directly edited or managed by the control property page to change the display property. The control properties in HTML are represented by the HTML segments,

<param name="BorderStyle" value="0"> or

<param name="Display" value>, and so forth.

The example shown below illustrates the necessary script to dynamically display a GUS page within the WebGUS control.

In this example, no display is associated with the WebGUS control when the control is first loaded.



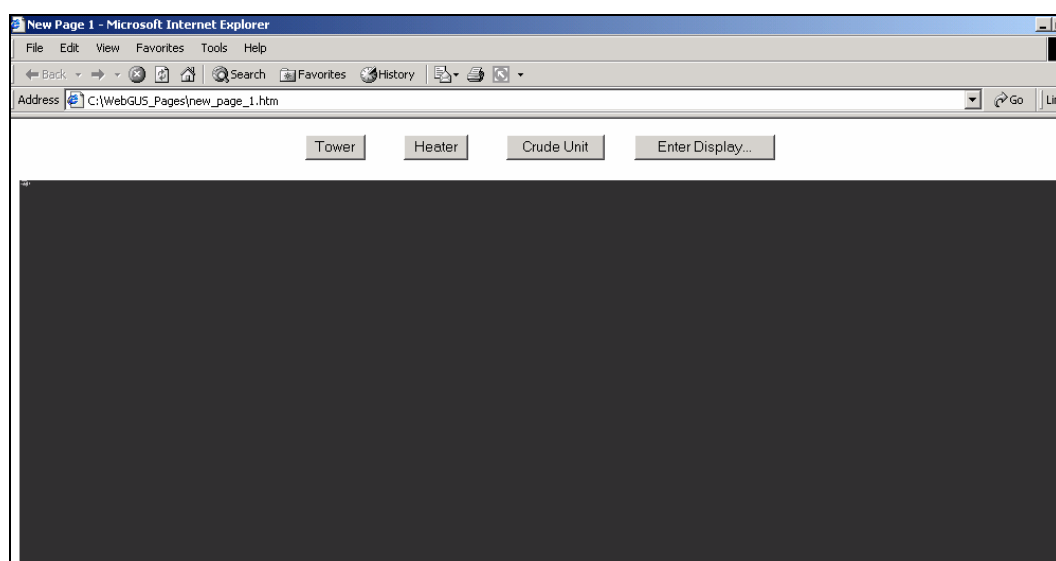
TIP

Insure that you 'Validate' your GUS display before you assign it to the WebGUS control.

3 Scripting WebGUS ActiveX Controls

3.1 Introduction to WebGUS ActiveX Control Scripting

Web Page with Embedded WebGUS Control



3 Scripting WebGUS ActiveX Controls

3.1 Introduction to WebGUS ActiveX Control Scripting

```
<html>

<head>
<meta http-equiv="Content-Language" content="en-us">
<meta http-equiv="Content-Type" content="text/html; charset=windows-1252">
<meta name="GENERATOR" content="Microsoft FrontPage 4.0">
<meta name="ProgId" content="FrontPage.Editor.Document">
<title>New Page 1</title>
</head>

<!--The script below loads a GUS display to the WebGUS Control depending on which
button is clicked on the web page. -->
<SCRIPT LANGUAGE="VBScript">

<!--

sub Btn1_onClick()
    WebGUS1.display = "Tower"
end sub

sub Btn2_onClick()
    WebGUS1.display = "Heater"
end sub

sub Btn3_onClick()
    WebGUS1.display = "Crude"
end sub
```

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3 Scripting WebGUS ActiveX Controls

3.1 Introduction to WebGUS ActiveX Control Scripting

```
<param name="Zoom" value="0">  
<param name="EnableRuntime" value="1">  
<param name="Keylock" value="-1">  
<param name="CommonGDDB" value="0">  
</object>  
</p>  
  
</body>  
  
</html>
```

3.2 Scripting WebGUS ActiveX Control Properties

WebGUS provides the following Properties. The container's scripting environment may access the properties via the WebGUS control's instance object name. Similarly, the scripting environment may handle the events raised by the WebGUS instantiation.



TIP

Links to scripting examples can be found in [WebGUS Help](#).

Properties provided by the web page container



ATTENTION

Refer to the appropriate container (FrontPage or Internet Explorer) documentation for information regarding extended properties added by the container to its embedded ActiveX controls.

Version

Data Type	Description	Usage	Default
	An Extended Property added by the hosting container.		

3 Scripting WebGUS ActiveX Controls

3.2 Scripting WebGUS ActiveX Control Properties

_ExtentX

Data Type	Description	Usage	Default
	An Extended Property added by the hosting container.		

_ExtentY

Data Type	Description	Usage	Default
	An Extended Property added by the hosting container.		

Properties provided by WebGUS

_StockProps



ATTENTION

_StockProps is a bit mask to indicate which standard ActiveX control properties the container can manipulate.

DO NOT modify this value.

Data Type	Description	Usage	Default
	An Extended Property added by the hosting container.		

BorderStyle

Data Type	Description	Usage	Default
integer	0 – no border 1 – show (thin line) border (1 pixel) The value can be changed at startup and anytime during control operation, immediately affecting the control.	Read-write	0

CommonGDDB

Data Type	Description	Usage	Default
BOOL	The CommonGDDB property indicates if GUS display that executes inside the WebGUS control uses container's instance of the global DISPDB or is it a new, separate instance. • CommonGDDB = FALSE. WebGUS at display startup will allocate new instance of global DISPDB and will initialize it per contents of the global DISPB indicated by the GDDB property. • CommonGDDB = TRUE. The display that executes in WebGUS control will use the global DISPDB that is associated with the container process (GUS display or Internet Explorer). This property is "fixed" for a newly invoked display after the <i>OnDisplayState</i> event is processed for the <i>_FileOpen</i> state. Subsequent changes to this property will not affect this display, but will affect subsequently invoked displays. The value stays in effect for the control's lifetime. This property value is not cleared after a display is invoked.	Read-write	FALSE

3 Scripting WebGUS ActiveX Controls

3.2 Scripting WebGUS ActiveX Control Properties

Description

Data Type	Description	Usage	Default
string	A string of up to 1000 characters, provided for the purpose of describing a particular instantiation of a WebGUS control. Can be set via the GUS container only. On the Property Page, the field is grayed when viewed from build-time containers other than GUS Display Builder.	Read-write	NULL

Display

Data Type	Description	Usage	Default
string	The GUS display file that is to be shown by the control. Changing this property value replaces current display. An empty string closes current display. The display name may be fully qualified or not. If not, the display file must be present in the display search path. If the display is not found, an error dialog is displayed indicating the given display was not found. On the Property Page, this field defines the first GUS display to be shown. If left blank, then no display is invoked when the control is first activated, and the area is rendered either gray or according to the inheritbackground property described below. Selecting the "Browse" button to the right of the display field enables the user to use the standard Windows File/Open dialog to browse to the desired display path	Read-write	NULL

DisplayParameter

Data Type	Description	Usage	Default
Variant	A GUS feature allows a parameter, of type variant, to be passed from the InvokeDisplay actor, which is part of the GUS scripting. This parameter is passed to the invoked display, which may then retrieve the parameter using the GUS script function called <i>GetDisplayParameter</i> The DisplayParameter property is provided to allow this parameter to be passed to displays invoked into the given WebGUS control. They must likewise use <i>GetDisplayParameter</i> to retrieve the parameter value This property is "fixed" for a newly invoked display after the <i>OnDisplayState</i> event is processed for the <i>_FileOpen</i> state. Subsequent changes to this property will not affect this display, but will affect subsequently invoked displays. This property value is cleared after each display invocation, in order to prevent inadvertently setting the parameter on subsequent display invocations	Write	NA

EnableRuntime

Data Type	Description	Usage	Default
BOOL	This property is for internal use during configuration. This property must be TRUE during runtime.	Read-write	TRUE

3 Scripting WebGUS ActiveX Controls

3.2 Scripting WebGUS ActiveX Control Properties

FastGroup

Data Type	Description	Usage	Default
BOOL	Allows activation (FastGroup = TRUE) and de-activation (FastGroup= FALSE) of the “fast group” scan updates (the FST group) on a running GUS display. Resets to FALSE during display startup For more information on fast-group functionality, please refer to <i>Optimizing GUS data Collection Groups</i> in the <i>GUS Display Authoring Tutorial</i>.	Read-write	FALSE

GDDB

Data Type	Description	Usage	Default
integer	Id of the global DISPDB used by the display executing in the WebGUS control. For more information on the DISPDB ID, see Display Database (DispDB) in the <i>GUS Display Scripting User's Guide</i>. Handling of the GDDB property depends on setting of the CommonGDDB property. - CommonGDDB = TRUE. the GDDB is a read-only property and indicates id of the global DISPDB that is being used by both WebGUS control and the container. - CommonGDDB = FALSE. the GDDB indicates id of the global DISPDB that's is used by this instance of the WebGUS control. While FALSE, container script is allowed to specify the value of the GDDB to force a specific global DISPDB as the new global DISPDB "initializer" for the next display. GDDB = -1 indicates global DISPDB that is used by the container. Value range: -1 or 1:20 This property is "fixed" for a newly invoked display after the <i>OnDisplayState</i> event is processed for the <i>_FileOpen</i> state. Subsequent changes to this property will not affect this display, but will affect subsequently invoked displays. This property value is not cleared after a display is invoked.	Read-write	-1

InheritBackground

Data Type	Description	Usage	Default
BOOL	Indicates if control's GUS display uses container background color. - InheritBackground = TRUE. The display background color is defined by the (ambient) Error! Reference source not found. Property. - InheritBackground = FALSE. The WebGUS display's predefined background color is retained. If no display is active in the WebGUS ActiveX control, the background is gray. The value can be changed at startup and anytime during control operation, immediately affecting current display.	Read-write	FALSE

Visible

Data Type	Description	Usage	Default
BOOL	Visibility in the GUS container at runtime. When TRUE, the WebGUS display (or control background if no display is set) is visible, including the border, if specified. When FALSE, the WebGUS display becomes invisible, thereby showing the container background including any underlying objects. Applies to the GUS container only. This property UI is grayed and checked (set TRUE) when used in non-GUS build-time containers.	Read-write	TRUE

Zoom

Data Type	Description	Usage	Default
integer	The zoom factor applied to the current and subsequent WebGUS displays. The default value (zero) affects the current and subsequently invoked WebGUS displays such that they are shown at their own <i>built-in</i> initial zoom setting. Legal values include: • -1 – “Zoom To Fit” • 0 – WebGUS display is shown at the display’s own built-in initial zoom factor. • 5 ... 1000 – 5% to 1000% zoom range. Any setting outside of these will result in no change to the property. Note: if display is shown in a regular zoom mode (i.e. no zoom to fit), then it is possible to have the vertical/horizontal scrollbars along the display <i>independently</i> of the Border setting. That is, it is possible to have scroll bars on the right/lower sides, but no border on the left/upper sides	Read-write	0

3.3 Scripting Keylock property for Web page

Keylock

Data Type	Description	Usage	Default
Int (-1...3)	This property is provided for use within a web page (Internet Explorer) only. The WebGUS control provides the “Keylock” property to explicitly select security key level for the WebGUS display. - VIEW_ONLY (= 0) - OPERATOR (= 1) - SUPERVISOR (= 2) - ENGINEER (= 3) The keylock selection applies to the particular instance of the WebGUS control and initializes keylock value during each display invocation in this control. Web page script may change value of the “Keylock” property while the display is running and then the change takes effect immediately. The <USE AMBIENT> value (= -1) is reserved to indicate that the container does not explicitly select security level for the WebGUS display. Other sources (such as internal GUS scripting actors and/or COP) do. In absence of other sources of the key level the WebGUS displays will default operation to the VIEW_ONLY level. On read, the “Keylock” property returns the value of the current security key level that the WebGUS display is operating at. NOTES: - Initial value of the “Keylock” property can be selected when embedding the control into a web page using the MS FrontPage	Read-write	-1

3 Scripting WebGUS ActiveX Controls

3.3 Scripting Keylock property for Web page

Data Type	Description	Usage	Default
	ActiveX control configuration "Parameters" tab. 2. Changing the value of "Keylock" (range 0 – 3) is equivalent to internally executing the KEY_CHG scripting actor. The "keylock" setting in range of 0 - 3 disables internal changes of the key level by the scripting actors. So that they can feel safer, when they set it to e.g. view-only : no actor that may be in the display will change it. 3. To allow write operations from the WebGUS display in an IE container, the value of the "Keylock" property must be set > 0. Additionally, the destination HCI/OPC server may have to be configured to allow Write operations. 4. In the GUS display container, the <USE_AMBIENT> is enforced at all times.		

3.4 Scripting WebGUS Control Methods



ATTENTION

There are undocumented methods that are reserved for internal use only and should not be invoked by a user.

CancelFunction

Method	Usage
Void CancelFunction()	To prevent execution of a GUS display scripting function and the following script, that is announced by the event to the container. Refer to the description of events later. Container's script may invoke the CancelFunction while processing a particular event.

SelectFile



ATTENTION

The EnableRuntime property must be set to False for this method to execute.

Method Name	Parameters	Usage
String SelectFile(String FileExtension, String FileDescription, String InitialDirectory)	FileExtension = The filter that the Open Dialog will apply (.HTM or .PCT). If the call specifies a file extension, the dialog filters on the extension, otherwise the dialog defaults the filter to "All Files". FileDescription - The description associated with the file extension. If the call specifies a file description with the file extension, the dialog integrates the file description and the file extension otherwise it ignores the file description. InitialDirectory - The directory that the Open Dialog will open first. If the function call specifies an initial directory, the dialog displays files in that initial directory, otherwise it defaults to the last directory the user opened.	This method is used when the user of the web page needs to be able to browse the file system in order to find an Environment file that they want to use in their WebGUS setup. It invokes the standard Microsoft File Open dialog. Returns an empty string if there is no selection, ie no fully qualified path and file name or, if the user Cancels the operation. See SelectFile for an example of the HTML code for this method.

3 Scripting WebGUS ActiveX Controls

3.4 Scripting WebGUS Control Methods

Refresh

Method	Usage
Refresh()	To force repaint of the GUS display, e.g. after changing a property of an internal GUS object that one wishes to be immediately reflected on-screen.

GetSVCcategory

Method	Usage
String GetSVCcategory (display name)	Returns the SafeView category as defined for the specified display. Display name must be either a fully qualified GUS display file name, or must be found within current display search path.

GetMouseX

Method	Usage
Integer GetMouseX ()	Returns current X position of the mouse in screen coordinates.

GetMouseY

Method	Usage
Integer GetMouseY ()	Returns current y position of the mouse in screen coordinates.

3.5 Scripting WebGUS ActiveX Control Events

WebGUS controls raise events that the container can then process. These events allow the container to intercept and do processing to either augment or replace the existing GUS display's script processing. Events can be raised in one of two ways.

1. Automatically – these events are raised for display state changes, and whenever existing scripted events actors are encountered.
2. Explicitly – in a modified GUS display script, by executing the *CustomEvent* actor. Below are the events that WebGUS controls can raise to the containing application.



TIP

[Links to scripting examples can be found in WebGUS Help.](#)

OnDisplayState

Event Name	Parameters	Usage
OnDisplayState	State = {"_FileOpen", "_StartupBegin", "_StartupComplete", "_ShutdownBegin", "_ShutdownComplete"} FileName = string, a fully qualified GUS display PCT file name. Obj = dispatch pointer to the GUS display object.	Reports current phase of the GUS display. Display invocation (or shutdown) is suspended while container processes the event. Note: the FileName and Obj parameters are not set (NULL) when this event is raised on the <i>FileOpen</i> and <i>_ShutdownComplete</i> states. <i>_FileOpen</i> fires before a display being opened has been read from disk. <i>_StartupBegin</i> fires after display has been read from the disk, before startup scripts are activated. <i>_StartupComplete</i> fires when all startup scripts have completed, before activating the <i>OnDataChange</i> scripts. <i>_ShutdownBegin</i> fires when display closes, before the shutdown scripts are activated. <i>_ShutdownComplete</i> fires when all shutdown scripts have completed. The <i>Obj</i> pointer can be used to set/get GUS display parameter(s). E.g. if "par_int" is a GUS display parameter, then use. <i>Obj.params.par_int</i> = <new value> to set parameter's value.

OnDisplayActor

Event Name	Parameters	Usage
OnDisplayActor	Parameters are specific to the Actor that raised the event.	WebGUS display scripts automatically raise the "OnDisplayActor" event before executing the indicated actor's function internally. Note that many actors behave differently (e.g., "No-op") by default when executing within a WebGUS display.



TIP

While processing the "OnDisplayActor" event, the container script may execute the `CancelFunction()` method on the WebGUS control to prevent further execution of the actor and any following script lines by the GUS display.

Refer to the OnDisplayActor list in the appendix for list of supported functions and specification of parameters.



ATTENTION

If a container script takes over execution of an event then the container must handle any errors that may result from servicing this event.

OnInvokeDisplay

Event Name	Parameters	Usage
OnInvokeDisplay	DisplayName = string Parameter requested in InvokeDisplay or SCHEM actors. WindowSpec = String, An optional second parameter that specifies a window specification name within a SafeView configuration to be used to manage the resulting display window. DisplayParameter = Variant, an optional parameter made available to the invoked display. See the DisplayParameter. Obj = dispatch pointer to the GUS object that is executing the script.	Generated by the standard GUS display scripting actors InvokeDisplay and SCHEM.



TIP

While processing the “OnDisplayActor” event, the container script may execute the `CancelFunction()` method on the WebGUS control to prevent further execution of the actor and any following script lines by the GUS display.

Refer to the OnDisplayActor list in the appendix for list of supported functions and specification of parameters.



ATTENTION

If a container script takes over execution of an event then the container must handle any errors that may result from servicing this event.

OnStandardEvent



TIP

The GUS UI scripting thread is suspended while the container processes the OnStandardEvent.

The event must be scripted in GUS on a particular object, otherwise the OnStandardEvent does not fire.

Example: Left-button click on a selectable object but without script will not fire the OnStandardEvent).

The intent is to replace existing GUS scripting, instead of providing one.

Container may indicate that the particular script event processing be skipped by executing the CancelFunction() method while processing the OnStandardEvent.

Event Name	Parameters	Usage
OnStandardEvent	Id = GUS scripting standard mouse-event name, prefixed with '_'. E.g. "_OnLButtonClick" or "_OnRButtonUp". Obj = dispatch pointer to the GUS object that is about to execute the script.	Event fires before a GUS standard <i>mouse-event</i> executes. Only GUS User Interface scripting (UI) thread events are reported. Is limited to the following GUS script events: OnLButtonDown OnLButtonUp OnLButtonClick OnLButtonDoubleClick OnRButtonDown OnRButtonUp OnRButtonClick OnRButtonDoubleClick

OnCustomEvent



ATTENTION

A particular GUS scripting thread (most probably: UserInterface script thread) is suspended while the container processes the event.

Container may indicate that the script processing be aborted by executing the `CancelFunction()` method while processing the `OnCustomEvent`.

Event Name	Parameters	Usage
OnCustomEvent	Id = string uniquely identifying the custom event. Event ids prefixed with '_' are reserved for Honeywell. Parameter = variant as passed by the user. Obj = dispatch pointer to the GUS object that is executing the script.	The event fires as the result of the GUS script built-in function: RaiseCustomEvent Id as string, Parameter as variant.

3.6 GUS Scripting Considerations

Checking GUS Display Environment

See the *GUS Display Scripting User's Guide* for more information on;

GetGUSMode,

RaiseCustomEvent, and

InWebGUSControl

Enumerations in the WebGUS environment

Some items may not work as expected in the WebGUS environment. For example, the expression,

dispdb.enm01.internal = lcn.point.parameter.internal may report a scripting error because the internal format of the enumeration parameter will not completely match the format that is expected by the WebGUS display runtime.

In GUS, the internal form of the standard/custom enumeration combines both the enumeration set number and a particular set member number.

Initialization of a Display DataBase enumeration requires establishing both a set number and a member number. This can be scripted by:

dispdb.enm01.internal = enum("SetID:member").

An HCI/OPC server may not include the enumeration set number. Also, execution of,

Lcn.point.parameter = enum("SetID:member") (or similar), may result in the HCI/OPC server stripping the SetID value.



ATTENTION

The HCI TPN Server supports enumerations ".internal" form, but does not include the set number when communicating with the client.

As a rule, it is preferable to use external name forms for the enumerations in GUS displays.

3.7 Example HTML Code

SelectFile

The example HTML code below illustrates the usage of SelectFile in an HTML page.

The file used below is named SelectedGUSDisplay.htm. The HTML page is a template for viewing GUS displays selected by browsing the User File System.

The Web page includes the WebGUS ActiveX control by describing it with an <OBJECT> tag. The description specifies its unique clsid name and a list of WebGUS specific parameters.

The Browse button script uses the name to reference WebGUS and call the WebGUS function **SelectFile**.

```
<HTML>
```

```
<HEAD>
```

```
<META NAME="GENERATOR" Content="Microsoft FrontPage 4.0">
```

```
<TITLE>WebGUS Selected GUS Display Sample</TITLE>
```

```
<SCRIPT language="vbscript" >
```

```
Sub btnBrowse_onclick
```

```
Dim strFilePath
```

```
strFilePath = wgMain.SelectFile( "pct", "GUS Displays", "" )
```

```
If (strFilePath <> "") Then
```

```
    wgMain.Display = strFilePath
```

```
End If
```

```
End Sub
```

```
</SCRIPT>

</HEAD>

<BODY bgcolor="white" text="black" link="mediumblue" vlink="darkgray"
alink="red">

    <table border="0" width="91%">
        <tr>
            <td width="15%">
                <INPUT language="vbscript" type="button"
value="Browse..." name="btnBrowse" style="position: relative; width: 130; height: 50">
            </td>

            <td width="85%">Select a GUS Display (.pct).
            </td>
        </tr>
    </table>

    <p>
        <OBJECT classid="clsid:5D9ADDE9-F63B-11D4-8FB0-00C04F54788C"
name="wgMain" width="990" height="550">
            <param name="_Version" value="65536">
            <param name="_ExtentX" value="20955">
            <param name="_ExtentY" value="11642">
            <param name="_StockProps" value="33">
```

3 Scripting WebGUS ActiveX Controls

3.7 Example HTML Code

```
<param name="BorderStyle" value="0">
<param name="Display" value>
<param name="InheritBackground" value="0">
<param name="Zoom" value="0">
<param name="EnableRuntime" value="-1">
<param name="Keylock" value="-1">
<param name="CommonGDDB" value="0">
</OBJECT>
</p>
</BODY>

</HTML>
```

4. Examples of Integrating WebGUS

4.1 Integrating WebGUS ActiveX control in a web page

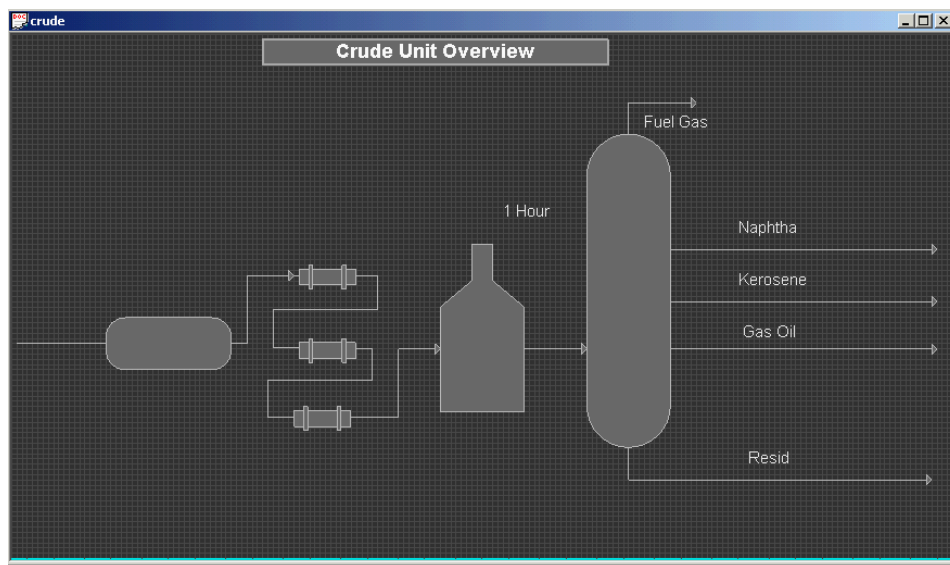
This section shows you how to create a web page that will host a GUS display using the WebGUS control.

Example GUS display

Shown below is an example of a GUS display. This could be a new display that you have created using GUS Display Builder or an existing display that is being used in your present operation.

You make a reference to this display via the WebGUS control Display property. Once referenced in the WebGUS ActiveX Control, the GUS display will be embedded in the WebGUS control.

Sample GUS Display

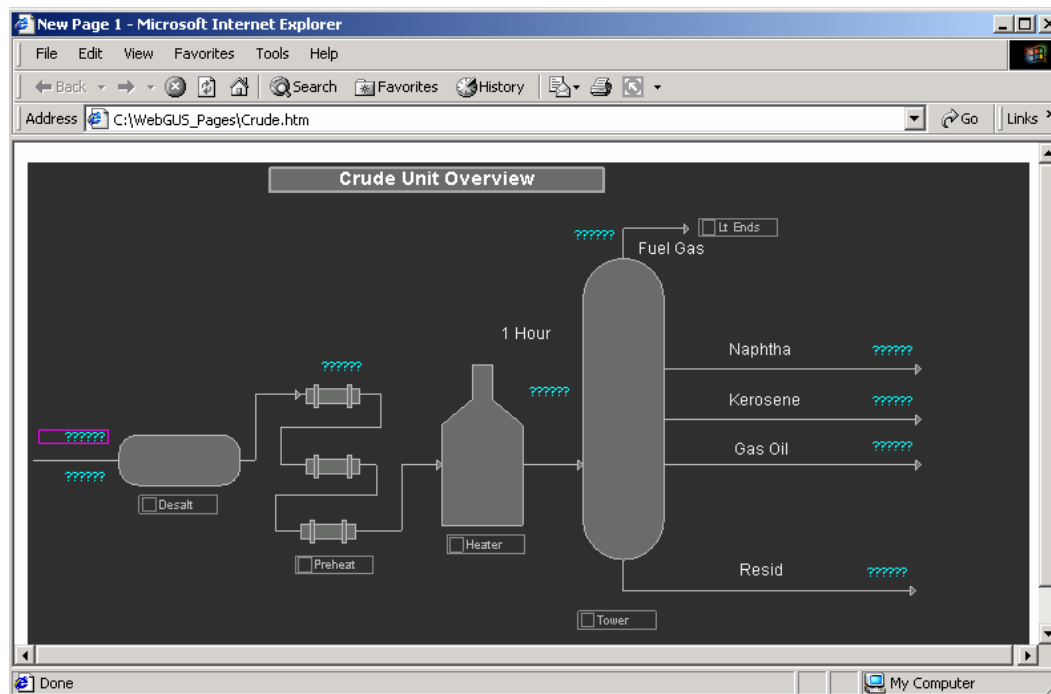


4 Examples of Integrating WebGUS

4.1 Integrating WebGUS ActiveX control in a web page

GUS display in the WebGUS control

The picture below illustrates the GUS display, as it would appear in the WebGUS control hosted by a web page.



Example HTML code

The code below illustrates the resulting code when a WebGUS control is added to the page (see Inserting WebGUS ActiveX Control into web page) and the Display property is set to display the example GUS control above (see Invoking WebGUS Property Page).



TIP

The HTML can be edited directly to add the display name to the line '<param name="Display" value="Crude.pct">' but the same can be accomplished, with more reliable results, by using the WebGUS Control property page that is exposed by the HTML editor, in this case FrontPage2000.

```
<html>
<head>
<meta http-equiv="Content-Type" content="text/html; charset=windows-1252">
<meta name="GENERATOR" content="Microsoft FrontPage 4.0">
<meta name="ProgId" content="FrontPage.Editor.Document">
<title>New Page 1</title>
</head>
<body>
<p>
<object classid="clsid:5D9ADDE9-F63B-11D4-8FB0-00C04F54788C" id="WebGUS1"
width="750" height="523">
  <param name="_Version" value="65536">
  <param name="_ExtentX" value="19844">
  <param name="_ExtentY" value="13838">
  <param name="_StockProps" value="33">
  <param name="BorderStyle" value="0">
  <param name="Display" value="Crude.pct">

```

4 Examples of Integrating WebGUS

4.1 Integrating WebGUS ActiveX control in a web page

```
<param name="InheritBackground" value="0">  
<param name="Zoom" value="0">  
<param name="EnableRuntime" value="1">  
<param name="Keylock" value="-1">  
<param name="CommonGDDB" value="0">  
</object>  
</p>  
</body>  
</html>
```

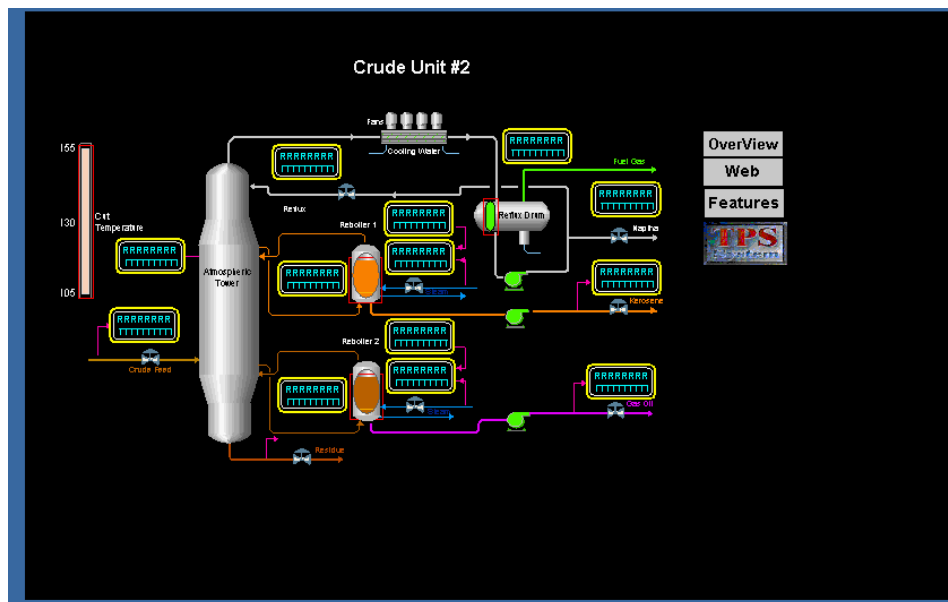
4.2 Integrating GUS displays using WebGUS in web page

This section shows you how to dynamically call up different GUS displays in the WebGUS control during run time within a web page.

The “Tower” button on the web page calls up a GUS display that is shown in the example below. The “Enter Display” button allows the user to type in a fully qualified path to an existing display on their system, or one that is accessible over a network that is connected to their system.

You can add more buttons with more linked GUS displays by repeating the button relevant HTML code that appears in the Example HTML code below.

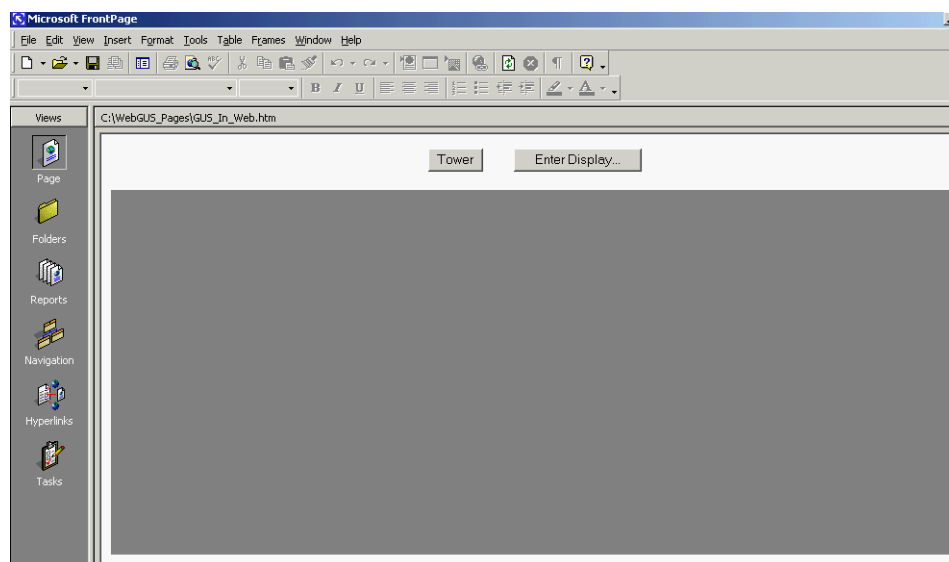
Example GUS display



4 Examples of Integrating WebGUS

4.2 Integrating GUS displays using WebGUS in web page

Example navigation buttons in web page



Example HTML code

```
<html>
<head>
<meta http-equiv="Content-Type" content="text/html; charset=windows-1252">
<meta name="GENERATOR" content="Microsoft FrontPage 4.0">
<meta name="ProgId" content="FrontPage.Editor.Document">
<title>GUS_in_Web</title>
</head>
<SCRIPT LANGUAGE="VBScript">
<!--
```

```
sub Btn1_onClick()  
    WebGUS1.display = "Tower_hci_phd.pct"  
end sub  
  
sub Btn2_onClick()  
    WebGUS1.display = inputbox("Display to load:")  
end sub  
  
-->  
</SCRIPT>
```

<body>

<p align="center">

```
<input type="button" value="Tower" name="Btn1">
```

[illegible]

<p>

<object classid="clsid:5D9ADDE9-F63B-11D4-8FB0-00C04F54788C" id="WebGUS1" width="899" height="451">

```
<param name="_Version" value="65536">
```

```
<param name="_ExtentX" value="23098">
```

```
<param name="_ExtentY" value="10028">
```

```
<param name="_StockProps" value="33">
```

4 Examples of Integrating WebGUS

4.2 Integrating GUS displays using WebGUS in web page

```
<param name="BorderStyle" value="0">
<param name="Display" value>
<param name="InheritBackground" value="0">
<param name="Zoom" value="0">
<param name="EnableRuntime" value="-1">
<param name="Keylock" value="-1">
<param name="CommonGDDB" value="0">
</object>
</p>
</body>
</html>
```

4.3 Integrating GUS display in a GUS display using WebGUS

This section shows you how to embed a GUS display into another GUS display (overlay) utilizing the WebGUS control.

Example GUS display in GUS display

Shown below is an example of a GUS display that contains another GUS display using the WebGUS control. See the Insert WebGUS ActiveX Control procedure to add a WebGUS control to an exiting GUS display.



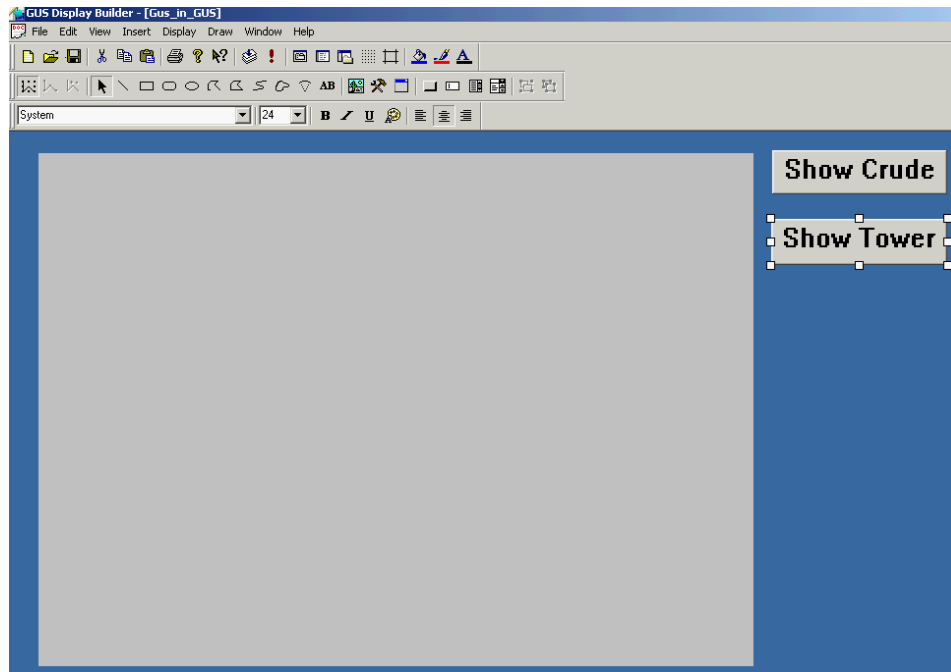
ATTENTION

You must first install and register the WebGUS ActiveX Control on your system before the control can be used to display GUS displays. See [Using the Environment Selection Section in the WebGUS Planning and Installation Guide](#).

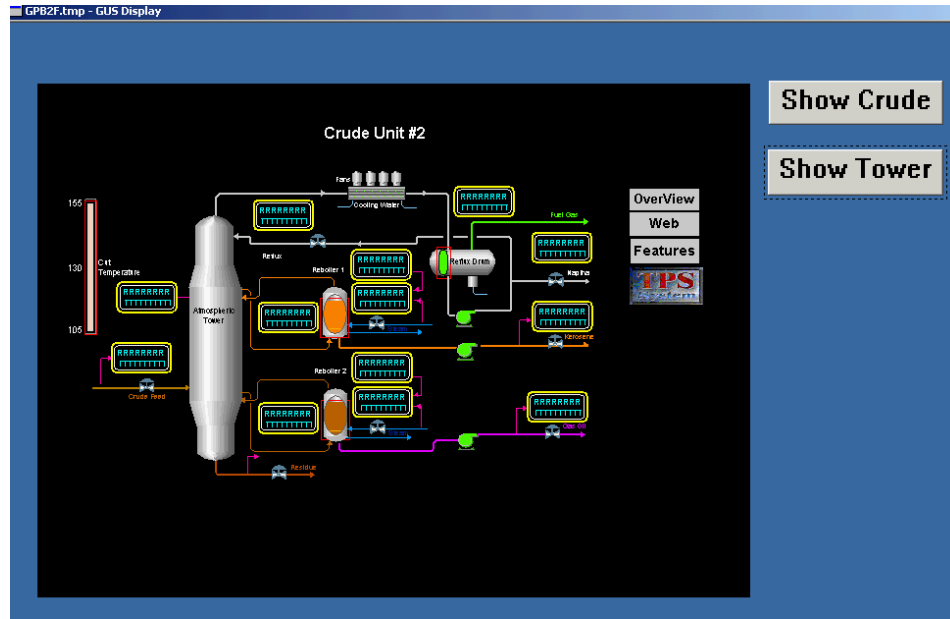
4 Examples of Integrating WebGUS

4.3 Integrating GUS display in a GUS display using WebGUS

Build Time GUS-in-GUS Using the WebGUS Control



Run Time GUS-in-GUS Using the WebGUS Control



Example display script

The script below is the script contained in the Show Tower button. This script loads the desired GUS display into the WebGUS control when the Show Tower button is clicked.

```
Sub OnLButtonClick()  
    oleobject1.display = "tower_hci_phd.pct"  
End Sub
```

4.4 Accessing GUS display parameters from a web page

Example display script

A web page script can drive new values into a WebGUS display parameter, thus causing the display to, for instance, show information relating to a currently selected tag name. To access display parameters, first save the *display* object, which is provided via the *_StartupBegin*, *_StartupComplete*, and *_ShutdownBegin* *OnDisplayState* state-change events, described in this document. Display parameters are then accessed via the display object's *params* property.



ATTENTION

The specific parameters available for a given display vary with each display; consult the displays for which you are interested in accessing parameters.

4.5 Examples of Accessing Global Database

Example GUS display in GUS display

The container, either a web page or a GUS display, includes a WebGUS ActiveX control, named WebGUS1.

The container might contain script such as;

```
WebGUS1.display = "a"
```

```
.....
```

```
.....
```

```
WebGUS1.display = "b".
```

The contents of the Global dispdb of "b" at startup will equal the contents of Global dispdb of "a";

If, CommonGDDB = TRUE – "b" will use the same instance unchanged (container's).

If CommonGDDB = FALSE – "b" will allocate a new instance and then copy from "a".

Example SafeView emulation

A web page is built to include two WebGUS controls, WebGUS1 and WebGUS2.

WebGUS1 has a script that executes the OnInvokeDisplay actor.

The web page has the following event handler code.

```
Sub WebGUS1.OnDisplayInvocation (DisplayName)
```

```
    WebGUS2.GDDB = WebGUS1.GDDB
```

```
    WebGUS2.Display = DisplayName
```

```
End Sub
```

WebGUS2 before displaying "a" will allocate new instance of global dispdb and initialize as copy-from global dispdb of the display that executes in WebGUS1.

4 Examples of Integrating WebGUS

4.5 Examples of Accessing Global Database

In the next example, the containing web page determines which HCI data server the GUS display is to utilize:

The web page scripting would be:

‘: set the data server name before display attempts to bind to the hci server

```
sub WebGUS1_OnDisplayState(State, FileName, Obj)
```

```
if State = “_StartupBegin” then
```

```
    Obj.params.hciserver = “TPN_Server55”
```

```
end if
```

```
end sub
```

The GUS Display scripting would be:

REM display binds to hci server at a point after _StartupBegin event has been processed

```
sub OnDisplayStartup()
```

```
    hci.s.bind display.params.hciserver
```

```
end
```

REM sometime later, the GUS display shows data from the container-specified hci server

```
sub OnDataChange()
```

```
    me.text = hci.s.A100.pv
```

```
end
```

On Line Help

More scripting examples, including examples of overlay emulation, can be found in WebGUS Help.

5. Running GUS displays using WebGUS

5.1 User Interface to GUS displays using WebGUS

GUS display in browser

The WebGUS runtime HI is essentially a GUS display running in a window that is a component of a container application window. This display executes as a separate gpb.exe process. If for any reason this gpb.exe process fails, a rectangle is presented in its place with the internationalizable string presented: “GUS Display Runtime Lost!”.

Intended use of desktop displays

CAUTION

Use of Internet Explorer as a container for mission critical control applications is not recommended or supported by Honeywell.

5.2 WebGUS ActiveX Control Runtime Functionality

Introduction to WebGUS ActiveX control at runtime

The WebGUS software provides a GUS runtime in the form of an ActiveX control embedded in an ActiveX container in “run mode”.

WebGUS runtime environments can differ

The WebGUS display runtime should be familiar to users who are familiar with GUS displays running on a standard GUS node, with the following exception

- With WebGUS, the GUS display is shown as a component piece of a larger “containing” display.

There are, however, some significant differences in the runtime behavior of certain aspects of GUS displays depending on the runtime environment used

5.3 Interpreting GUS, Desktop, GUS using WebGUS Runtime Behaviors

The tables in this section contrast the differences between the following three-runtime environments;

- Standard, LCNP-connected GUS, as familiar to users of GUS displays in control room environments.
- WebGUS ActiveX as used on a non LCNP-connected desktop.
- WebGUS ActiveX as used on an LCNP-connected GUS platform. The WebGUS ActiveX in this case may be embedded in either a GUS display or in a web page in Internet Explorer.

LCN Dependent actors behavior

LCNP-Connected GUS Display Behavior	Non LCNP-Connected Desktop WebGUS Display Behavior	LCNP Connected WebGUS Display Behavior
Actors execute as documented in the <i>GUS Display Scripting User's Guide</i> .	As each existing GUS display actor is encountered at runtime in a WebGUS display, the <i>OnDisplayActor</i> event is automatically raised by the WebGUS control to the container, allowing the container an opportunity to replace the processing normally handled by the actor. See <i>OnDisplayActor</i> for information on the <i>r</i> event. Note that aside from user-defined container-side processing, many actors result in a simple "No-op" behavior.	Actors execute as documented in the <i>GUS Display Scripting User's Guide</i> .

Collectors behavior

LCNP-Connected GUS Display Behavior	Non LCNP-Connected Desktop WebGUS Display Behavior	LCNP Connected WebGUS Display Behavior
Collectors execute as documented in the <i>GUS Display Scripting User's Guide</i> .	When encountered during runtime in a WebGUS display, collectors return a default value. For information on the specific expected default values for collectors, see the GUS Display Scripting Guide.	Collectors execute as documented in the GUS Display Scripting Guide.

LCN Dependent ActiveX controls behavior

The table addresses the Trend, AlarmWindow, and Faceplate controls.

LCNP-Connected GUS Display Behavior	Non LCNP-Connected Desktop WebGUS Display Behavior	LCNP Connected WebGUS Display Behavior
These controls behave as documented in the <i>GUS Display Scripting User's Guide</i> .	A "red-x" error status indication is presented, similar to when a connection/configuration error occurs at runtime for these controls. (See .) Note that WebGUS automatically re-creates the "red-x" error indication in the absence of installed Trend, AlarmWindow or Faceplate controls. (They are not included with the WebGUS product software.).	These controls behave as documented in the GUS Display Scripting Guide.

5 Running GUS displays using WebGUS

5.3 Interpreting GUS, Desktop, GUS using WebGUS Runtime Behaviors

LCN Data Redirection using WebGUS behavior

LCNP-Connected GUS Display Behavior	Non LCNP-Connected Desktop WebGUS Display Behavior	LCNP Connected WebGUS Display Behavior
Does not apply. No data redirection occurs. Built-in displays' data references are used.	Operates as described in the section of the WebGUS Planning and Administration Guide, to automatically convert data references into surrogate HCI/OPC data references. Based on either default or customized configuration, all LCN name references found in WebGUS displays, including those in scripts, are redirected to go to the configured HCI/OPC server or servers.	Does not operate. No data redirection occurs. Built-in displays' data references are used.

Right mouse behavior

LCNP-Connected GUS Display Behavior	Non LCNP-Connected Desktop WebGUS Display Behavior	LCNP Connected WebGUS Display Behavior
Operates as documented in the <i>GUS Display Scripting User's Guide</i> (a pop-up menu is presented).	When executing within a GUS container, the container display provides the same pop-up menu as a normal GUS display. No action occurs when executing in a non-GUS container such as Internet Explorer.	Same as desktop behavior.

Concurrent display behavior

LCNP-Connected GUS Display Behavior	Non LCNP-Connected Desktop WebGUS Display Behavior	LCNP Connected WebGUS Display Behavior
Operates as documented in the <i>GUS Display Scripting User's Guide</i> . A maximum concurrent number of displays (e.g., 4 or 8 displays) are enforced by the picture manager.	A maximum of 16 concurrent WebGUS displays is allowed. Of course, system resources, including those of the surrogate HCI/OPI data servers, must be considered as the number of concurrent WebGUS displays increases.	Because non-desktop WebGUS displays utilize the same resources as standard GUS displays (e.g., HOPC data server resources), the number of concurrent WebGUS and GUS displays is limited per the available GUS node resources. This is enforced by the Picture Manager and Safeview.

Title and status bar

LCNP-Connected GUS Display Behavior	Non LCNP-Connected Desktop WebGUS Display Behavior	LCNP Connected WebGUS Display Behavior
Operates as documented in the <i>GUS Display Scripting User's Guide</i> .	Title and status bar not shown.	Title and status bar not shown.

5 Running GUS displays using WebGUS

5.3 Interpreting GUS, Desktop, GUS using WebGUS Runtime Behaviors

Display invocation behavior

LCNP-Connected GUS Display Behavior	Non LCNP-Connected Desktop WebGUS Display Behavior	LCNP Connected WebGUS Display Behavior
Invokes a GUS display using a separate gpb.exe process. Behavior depends on node configuration.	Replaces the current GUS display shown by the WebGUS ActiveX control with a given GUS display. An associated event, called <i>OnInvokeDisplay</i> , is raised allowing container to process and/or cancel a standard GUS <i>InvokeDisplay</i> script. See <i>OnInvokeDisplay</i> for information on the <i>OnInvokeDisplay</i> event	Same as desktop-WebGUS behavior

ACK key behavior

LCNP-Connected GUS Display Behavior	Non LCNP-Connected Desktop WebGUS Display Behavior	LCNP Connected WebGUS Display Behavior
Operates as documented in the <i>GUS Display Scripting User's Guide</i> .	The ACK propagates down over all contained WebGUS displays. See <i>ACK Behavior</i> in the GUS Display Scripting Users Guide. There is no direct acknowledgement of process alarms available from a desktop WebGUS display. When an ACK key-press indication occurs, a scripted ACK actor function executes as documented in see <i>ACK</i> under <i>ACKSTAT()</i> of the GUS Display Scripting Guide.	

5 Running GUS displays using WebGUS

5.3 Interpreting GUS, Desktop, GUS using WebGUS Runtime Behaviors

IKB Key handling behavior

The table below relates to the OnAssoc, OnClear, OnConfirm, OnDetail, OnDisplayBack, OnDisplayForward, OnPageBack, OnPageForward, Event Response, and PMK keys.

LCNP-Connected GUS Display Behavior	Non LCNP-Connected Desktop WebGUS Display Behavior	LCNP Connected WebGUS Display Behavior
Operates as documented in the GUS Display Scripting Guide. Additionally, QKEY will raise the DisplayActor event.	These key events can be simulated via the GUS QUE_KEY actor. When these IKB key events occur, the OnDisplayActor event for QKEY is raised. If not handled by a WebGUS script and not canceled by the container, then the specific key is forwarded to the container. In that case, only the GUS container will process the key message, as documented in the GUS Display Scripting Guide.	

PMK behavior

The PMK object may still be used in Desktop WebGUS controls, but requests to Raise, Lower, FastRaise, or FastLower, will generate error 6 PMKERR_RAMPFAILED (“Fail to obtain Internal Ramping Data”).

5.4 Behavior of Actors, Collectors, DDB in WebGUS

Job aid to Actors for WebGUS

The following table contains links to the GUS Display Scripting Guide where definitions and behavior of Actors can be found.



ATTENTION

For a complete description of Actors, along with syntax and examples, see the *Actor's Manual*

For a list of ...	See ...
Display Actors	<i>Display Actors</i> in the GUS Display Scripting Guide.
Operator Input Actors	<i>Operator Input Actors</i> in the GUS Display Scripting Guide.
Keyboard Event Actors	<i>Keyboard Event Actors</i> in the GUS Display Scripting Guide.
Store to DDB Actors	<i>Store to DDB Actors</i> in the GUS Display Scripting Guide.
Read data and Store into DDB Actor	Read data and Store into DDB Actor in the GUS Display Scripting Guide.
Read from DDB Actors (DispDB)	Read from DDB Actors (DispDB) in the GUS Display Scripting Guide.
Store to DDB Actors (DispDB)	Store to DDB Actors (DispDB) in the GUS Display Scripting Guide.
Read data and Store into System Actor	Read data and Store into System Actor in the GUS Display Scripting Guide.
Read from System DDB Actors	Read from System DDB Actors in the GUS Display Scripting Guide.
Range Check Actors	Range Check Actors in the GUS Display Scripting Guide.
General Actors	General Actors in the GUS Display Scripting Guide.

5 Running GUS displays using WebGUS

5.4 Behavior of Actors, Collectors, DDB in WebGUS

For a list of ...	See ...
Trend Actors	Trend Actors in the GUS Display Scripting Guide.
Arithmetic Actors	Arithmetic Actors in the GUS Display Scripting Guide.
Logical Actors	Logical Actors in the GUS Display Scripting Guide.
Comparative Actors	Comparative Actors in the GUS Display Scripting Guide.
Conditional Actors	Conditional Actors in the GUS Display Scripting Guide.
Conversion Actors	Conversion Actors in the GUS Display Scripting Guide.
Keylock Actors	Keylock Actors in the GUS Display Scripting Guide.

Job aid to Collectors for WebGUS

The following table contains links to the GUS Display Scripting Guide where definitions and behavior of Collectors can be found.



ATTENTION

For a description of how to use the `Collector()` function, see [Access to TPS Network Collector Data](#) in the GUS Display Scripting Guide.

For a list of ...	See ...
Alarm Collectors	<i>Alarm Collectors</i> in the GUS Display Scripting Guide.
History Collectors	<i>History Collectors</i> in the GUS Display Scripting Guide.

\$KEYLEVEL



TIP

The current KeyLevel can be obtained using a DispDB variable or the Collector shown below.

Collector Name	GUS Node Script Behavior	Desktop WebGUS Behavior
\$KEYLEVEL	Returns the state of the GUS Display Keylock Level. Collector("\$KEYLEVEL") will return one of the following values: VIEW, OPR, SUP, and ENGR	Same

5 Running GUS displays using WebGUS

5.4 Behavior of Actors, Collectors, DDB in WebGUS

Job aid to DDB for WebGUS

The following table contains links to the GUS Display Scripting Guide where definitions and behavior for Display Data Base can be found.

For a list of ...	See ...
Display Data Base	<i>Display Data Base</i>

5.5 Interpreting WebGUS Performance

Performance at invocation



ATTENTION

The performance of WebGUS is significantly affected by the hosting container. For example;

Multi-WebGUS displays experience a delay in redrawing when the containing web page is maximized during the time that the WebGUS displays are loading.

The display call up performance for a WebGUS display is greater than a standard GUS node display invocation, by either;

1. The amount of time it takes the operating system to start up the associated gpb.exe process, or if a display is already executing within a WebGUS control or,
2. The amount of time it takes for the existing display to complete its shutdown processing.

For initial WebGUS display invocation, this can be summarized as the sum of CT + AT, where;

CT = Call up time of the display when rendered by GUS node on a computer of equal resources and,

AT = Activation Time of the WebGUS control's associated gpb.exe process. The Activation time of the gpb.exe process should not exceed the activation time of a Microsoft Office 2000 application such as Word, within the same container, executing in the same machine as the WebGUS ActiveX control, and loading a document located on the same machine as the GUS display.

5 Running GUS displays using WebGUS

5.5 Interpreting WebGUS Performance

Display replacement performance

For subsequent display invocation within the same WebGUS ActiveX control, i.e., display replacement, the performance can be summarized as;

Replacement Time = CT + ST where;

CT = Call up time of the display when rendered by GUS node on a computer of equal resources and,

ST = Shutdown Time of the currently executing WebGUS display.

This is because the same gpb process that is used for the existing display is used again for the new, replacing display.

Display execution performance

A WebGUS display will provide the same performance as provided when that GUS display is executed in a GUS node, assuming a computer with similar resources.

Specifically, the following key activities will perform as quickly on WebGUS display as those same displays on GUS nodes, with the exception that OPC servers may add data access delay times. That is, the activities below will be equivalent on WebGUS displays compared to GUS node displays using HCI/OPC data access to retrieve data from similarly configured HCI/OPC servers.

- Display Update
- Time to access data in response to user initiated events
- Display Shut down

Display scenarios to avoid



ATTENTION

Avoid embedding a WebGUS ActiveX control within itself, resulting in a “hall-of-mirrors” recursion effect. This could have negative impact on performance due to the recursion consuming too many system resources.

5.6 Troubleshooting WebGUS

Some behaviors are expected with WebGUS

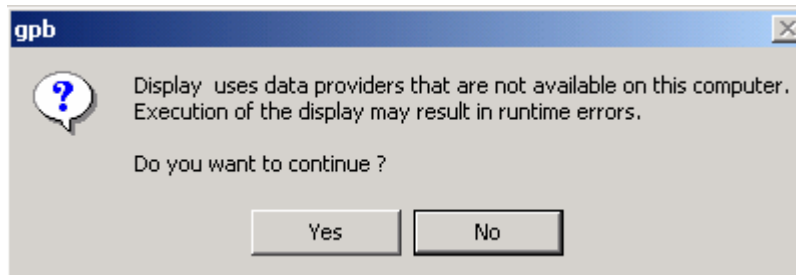
The WebGUS runtime HI is essentially a GUS display running in a window that is a component of a container application window. This display executes as a separate gpb.exe process. If for any reason this gpb.exe process fails, a rectangle is presented in its place with the internationalizable string presented: “GUS Display Runtime Lost!”.

Typical errors that may occur while using WebGUS

The sections below identify some of the typical errors that may occur, the indications that can expect, i.e. error message, and some of the actions that you can take to diagnose the error.

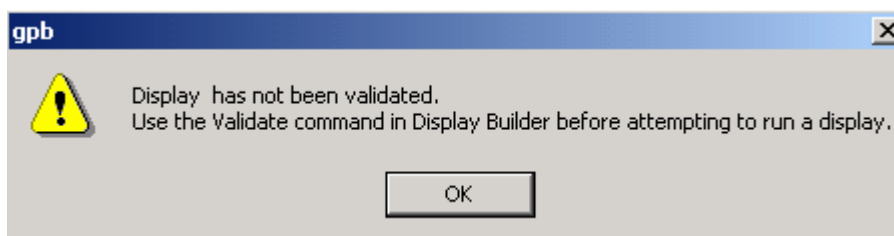
Communications with Data References

This error occurs when communications is lost with Redirection Server. The error message that you most likely will see is:



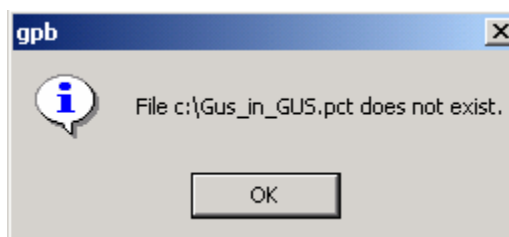
Validate Display Error

This error occurs when a WebGUS control tries to load a WebGUS display that has not been validated within GUS Display Builder. The error message that you most likely will see is:



Can't Locate Requested Display

The message will display the file that your WebGUS control is trying to access. The problem could be that the file is not in the location identified or, the location of the file (directory structure) is not identified in the System Path variable.



5 Running GUS displays using WebGUS

5.6 Troubleshooting WebGUS

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